Chapter 1

1.2 Absolute Value Inequalities

Definition of Absolute Value

The absolute value of a real number x, which is denoted by the symbol $|x|$, defined by

$$|x| = \begin{cases} x, & x \geq 0 \\ -x & x < 0 \end{cases}$$

Another definition of $|a| = \sqrt{(a)^2}$ (**Method 2**)

There are 2 cases involved when solving absolute value inequalities.

Case 1:

If $|x| < a$, then $-a < x < a$
If $|x| \leq a$, then $-a \leq x \leq a$

Case 2:

If $|x| > a$, then $x > a$ or $x < -a$
If $|x| \geq a$, then $x \geq a$ or $x \leq -a$

Example

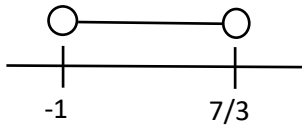
Determine the possible value of x if $|2 - 3x| < 5$. Give your answer on real number line.

Solution

This is Case 1 problem:

$-5 < 2 - 3x < 5$
$-5 - 2 < 2 - 3x - 2 < 5 - 2$
$-7 < -3x < 3$
$\frac{-7}{-3} > x > \frac{3}{-3}$ (Change the sign because it is divided by negative number)
$\frac{7}{3} > x > -1$
$-1 < x < \frac{7}{3}$

The real number line is:



Method 2

$$\sqrt{(2-3x)^2} < 5$$

$$\left[\sqrt{(2-3x)^2}\right]^2 < (5)^2$$

$$(2-3x)^2 < 25$$

$$4 - 12x + 9x^2 < 25$$

$$9x^2 - 12x - 21 < 0$$

$$(\div 3):\ 3x^2 - 4x - 7 < 0$$

$$(3x - 7)(x + 1) < 0$$

Critical values: $3x - 7 = 0$ or $x + 1 = 0$

$$x = \frac{7}{3} \text{ or } x = -1$$

Critical line:

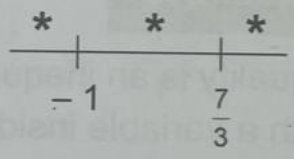
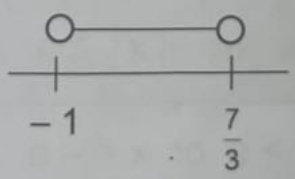


Table:

Interval	Test Value	Sign of $(3x - 7)(x + 1) < 0$
$(-\infty, -1)$	-5	$(-)(-) = +$
$\left(-1, \frac{7}{3}\right)$	0	$(-)(+) = -$ √
$\left(\frac{7}{3}, \infty\right)$	5	$(+)(+) = +$

Real number line:



Example

Determine the possible values of p if $6 - |2p - 5| < 4$. Give your answer in interval notation and real number line.

Solution

$$
\begin{aligned}
6 - |2p - 5| &< 4 \\
2 &< |2p - 5| \\
|2p - 5| &> 2 \\
2p - 5 > 2 \quad &\text{or} \quad 2p - 5 < -2 \\
2p > 7 \quad &\text{or} \quad 2p < 3 \\
p > \frac{7}{2} \quad &\text{or} \quad p < \frac{3}{2}
\end{aligned}
$$

Interval form : $\left(-\infty, \frac{3}{2}\right) \cup \left(\frac{7}{2}, \infty\right)$

Real number line :

